

COMP9020 23T1

Week 10

Expected Values, Course Review

- Textbook (R & W) - Ch. 9, Sec. 9.2-9.4
- Problem set week 10

Definition

An (integer) **random variable** is a function from Ω to $\mathbb{Z}$.
In other words, it associates a number value with every outcome.

Random variables are often denoted by $X, Y, Z, \dots$

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Expectation

Example

Random variable $X_s \stackrel{\text{def}}{=} \text{sum of rolling two dice}$

$$\Omega = \{(1, 1), (1, 2), \dots, (6, 6)\}$$

$$X_s((1, 1)) = 2 \quad X_s((1, 2)) = 3 = X_s((2, 1)) \quad \dots$$

Example

9.3.3 Buy one lottery ticket for \$1. The only prize is \$1M.

$$\Omega = \{\text{win}, \text{lose}\} \quad X_L(\text{win}) = \$999,999 \quad X_L(\text{lose}) = -\$1$$

Definition

The **expected value** (often called “expectation” or “average”) of a random variable X is

$$E(X) = \sum_{k \in \mathbb{Z}} P(X = k) \cdot k$$

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Example

The expected sum when rolling two dice is

$$E(X_s) = \frac{1}{36} \cdot 2 + \frac{2}{36} \cdot 3 + \dots + \frac{6}{36} \cdot 7 + \dots + \frac{1}{36} \cdot 12 = 7$$

Example

9.3.3 Buy one lottery ticket for \$1. The only prize is \$1M. Each ticket has probability $6 \cdot 10^{-7}$ of winning.

$$E(X_L) = 6 \cdot 10^{-7} \cdot \$999,999 + (1 - 6 \cdot 10^{-7}) \cdot -\$1 = -\$0.4$$

NB

Expectation is a truly universal concept; it is the basis of all decision making, of estimating gains and losses, in all actions under risk. Historically, a rudimentary concept of expected value arose long before the notion of probability.

Theorem (linearity of expected value)

$$E(X + Y) = E(X) + E(Y)$$

$$E(c \cdot X) = c \cdot E(X)$$

Example

The expected sum when rolling two dice can be computed as

$$E(X_s) = E(X_1) + E(X_2) = 3.5 + 3.5 = 7$$

since $E(X_i) = \frac{1}{6} \cdot 1 + \frac{1}{6} \cdot 2 + \dots + \frac{1}{6} \cdot 6$, for each die X_i

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Example

$E(S_n)$, where $S_n \stackrel{\text{def}}{=} |\text{no. of HEADS in } n \text{ tosses}|$

- 'hard way'

$$E(S_n) = \sum_{k=0}^n P(S_n = k) \cdot k = \sum_{k=0}^n \frac{1}{2^n} \binom{n}{k} \cdot k$$

since there are $\binom{n}{k}$ sequences of n tosses with k HEADS, and each sequence has the probability $\frac{1}{2^n}$

$$= \frac{1}{2^n} \sum_{k=1}^n \frac{n}{k} \binom{n-1}{k-1} k = \frac{n}{2^n} \sum_{k=0}^{n-1} \binom{n-1}{k} = \frac{n}{2^n} \cdot 2^{n-1} = \frac{n}{2}$$

using the 'binomial identity' $\sum_{k=0}^n \binom{n}{k} = 2^n$

- 'easy way'

$$E(S_n) = E(S_1^1 + \dots + S_1^n) = \sum_{i=1}^n E(S_1^i) = nE(S_1) = n \cdot \frac{1}{2}$$

Note: $S_n \stackrel{\text{def}}{=} |\text{HEADS in } n \text{ tosses}|$ while each $S_1^i \stackrel{\text{def}}{=} |\text{HEADS in 1 toss}|$

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NB

If $X_1, X_2, \dots, X_n$ are independent, identically distributed random variables, then $E(X_1 + X_2 + \dots + X_n)$ happens to be the same as $E(nX_1)$, but these are very different random variables.

9

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Example

You face a quiz consisting of six true/false questions, and your plan is to guess the answer to each question (randomly, with probability 0.5 of being right). There are no negative marks, and answering four or more questions correctly suffices to pass. What is the probability of passing and what is the expected score?

To pass you would need four, five or six correct guesses. Therefore,

$$p(\text{pass}) = \frac{\binom{6}{4} + \binom{6}{5} + \binom{6}{6}}{64} = \frac{15 + 6 + 1}{64} \approx 34\%$$

The expected score from a single question is 0.5, as there is no penalty for errors. For six questions the expected value is $6 \cdot 0.5 = 3$

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Exercise

9.3.7

An urn has $m + n = 10$ marbles, $m \geq 0$ red and $n \geq 0$ blue.
7 marbles selected at random without replacement.
What is the expected number of red marbles drawn?

$$\frac{\binom{m}{0}\binom{n}{7}}{\binom{10}{7}} \cdot 0 + \frac{\binom{m}{1}\binom{n}{6}}{\binom{10}{7}} \cdot 1 + \frac{\binom{m}{2}\binom{n}{5}}{\binom{10}{7}} \cdot 2 + \dots + \frac{\binom{m}{7}\binom{n}{0}}{\binom{10}{7}} \cdot 7$$

e.g.

$$\begin{aligned} & \frac{\binom{5}{2}\binom{5}{5}}{\binom{10}{7}} \cdot 2 + \frac{\binom{5}{3}\binom{5}{4}}{\binom{10}{7}} \cdot 3 + \frac{\binom{5}{4}\binom{5}{3}}{\binom{10}{7}} \cdot 4 + \frac{\binom{5}{5}\binom{5}{2}}{\binom{10}{7}} \cdot 5 \\ &= \frac{10}{120} \cdot 2 + \frac{50}{120} \cdot 3 + \frac{50}{120} \cdot 4 + \frac{10}{120} \cdot 5 = \frac{420}{120} = 3.5 \end{aligned}$$

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Example

Find the average waiting time for the first HEAD, with no upper bound on the 'duration' (one allows for all possible sequences of tosses, regardless of how many times TAILS occur initially).

$$\begin{aligned} A = E(X_w) &= \sum_{k=1}^{\infty} k \cdot P(X_w = k) = \sum_{k=1}^{\infty} k \frac{1}{2^k} \\ &= \frac{1}{2^1} + \frac{2}{2^2} + \frac{3}{2^3} + \dots \end{aligned}$$

This can be evaluated by breaking the sum into a sequence of geometric progressions

$$\begin{aligned} & \frac{1}{2} + \frac{2}{2^2} + \frac{3}{2^3} + \dots \\ &= \left(\frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3} + \dots \right) + \left(\frac{1}{2^2} + \frac{1}{2^3} + \dots \right) + \left(\frac{1}{2^3} + \dots \right) + \dots \\ &= 1 + \frac{1}{2} + \frac{1}{2^2} + \dots = 2 \end{aligned}$$

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There is also a recursive 'trick' for solving the sum

$$A = \sum_{k=1}^{\infty} \frac{k}{2^k} = \sum_{k=1}^{\infty} \frac{k-1}{2^k} + \sum_{k=1}^{\infty} \frac{1}{2^k} = \frac{1}{2} \sum_{k=1}^{\infty} \frac{k-1}{2^{k-1}} + 1 = \frac{1}{2}A + 1$$

Now $A = \frac{A}{2} + 1$ and $A = 2$

NB

A much simpler but equally valid argument is that you expect 'half' a HEAD in 1 toss, so you ought to get a 'whole' HEAD in 2 tosses.

Theorem

The average number of trials needed to see an event with probability p is $A = \frac{1}{p}$.

$$\begin{aligned} \text{Proof: } A &= \sum_{k=1}^{\infty} k \cdot p \cdot (1-p)^{k-1} \\ &= \sum_{k=1}^{\infty} (k-1) \cdot p \cdot (1-p)^{k-1} + p \cdot \sum_{k=0}^{\infty} (1-p)^k \\ &= \sum_{k=1}^{\infty} k \cdot p \cdot (1-p)^k + p \cdot \frac{1}{p} \\ &= (1-p) \cdot A + 1 \end{aligned}$$

Exercise

9.4.12 A die is rolled until the first 4 appears. What is the expected waiting time?

$P(\text{roll } 4) = \frac{1}{6}$ hence $E(\text{no. of rolls until first } 4) = 6$

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Example

To find an object $\mathcal{X}$ in an unsorted list L of elements, one needs to search linearly through L . Let the probability of $\mathcal{X} \in L$ be p , hence there is $1-p$ likelihood of $\mathcal{X}$ being absent altogether. Find the expected number of comparison operations.

If the element is in the list, then the number of comparisons averages to $\frac{1}{n}(1 + \dots + n)$; if absent we need n comparisons. The first case has probability p , the second $1-p$. Combining these we find

$$E_n = p \frac{1 + \dots + n}{n} + (1-p)n = p \frac{n+1}{2} + (1-p)n = (1-\frac{p}{2})n + \frac{p}{2}$$

As one would expect, increasing p leads to a lower expected number E_n .

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As one would expect, increasing p leads to a lower expected number E_n .

One may expect that this would indicate a practical rule — that high probability of success might lead to a high expected value. Unfortunately this is *not* the case in a great many practical situations.

Many lottery advertisements claim that buying more tickets leads to better expected results — and indeed, obviously you will have more potentially winning tickets. However, the expected value *decreases* when the number of tickets is increased.

As an example, let us consider a punter placing bets on a roulette (outcomes: $0, 1 \dots 36$). Tired of losing, he decides to place \$1 on 24 'ordinary' numbers $a_1 < a_2 < \dots < a_{24}$, selected from among 1 to 36.

His probability of winning is high indeed — $\frac{24}{37} \approx 65\%$; he scores on any of his choices, and loses only on the remaining thirteen numbers.

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But what about his performance?

- If one of his numbers comes up, say a_i , he wins \$35 from the bet on that number and loses \$23 from the bets on the remaining numbers, thus collecting \$12. This happens with probability $p = \frac{24}{37}$.
- With probability $q = \frac{13}{37}$ none of his numbers appears, leading to loss of \$24.

The expected result

$$p \cdot \$12 - q \cdot \$24 = \$12 \frac{24}{37} - \$24 \frac{13}{37} = -\$ \frac{24}{37} \approx -65\text{¢}$$

Many so-called 'winning systems' that purport to offer a winning strategy do something akin — they provide a scheme for frequent relatively moderate wins, but at the cost of an occasional very big loss.

It turns out (it is a formal theorem) that there can be *no system* that converts an 'unfair' game into a 'fair' one. In the language of decision theory, 'unfair' denotes a game whose individual bets have negative expectation.

It can be easily checked that any individual bets on roulette, on lottery tickets or on just about any commercially offered game have negative expected value.

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Standard Deviation and Variance

Definition

For random variable X with expected value (or: **mean**) $\mu = E(X)$, the **standard deviation** of X is

$$\sigma = \sqrt{E((X - \mu)^2)}$$

and the **variance** of X is

$$\sigma^2$$

Standard deviation and variance measure how spread out the values of a random variable are. The smaller σ^2 the more confident we can be that $X(\omega)$ is close to $E(X)$, for a randomly selected ω .

NB

The variance can be calculated as $E((X - \mu)^2) = E(X^2) - \mu^2$

Example

Random variable $X_d \stackrel{\text{def}}{=} \text{value of a rolled die}$

$$\mu = E(X_d) = 3.5$$

$$E(X_d^2) = \frac{1}{6} \cdot 1 + \frac{1}{6} \cdot 4 + \frac{1}{6} \cdot 9 + \frac{1}{6} \cdot 16 + \frac{1}{6} \cdot 25 + \frac{1}{6} \cdot 36 = \frac{91}{6}$$

$$\text{Hence, } \sigma^2 = E(X_d^2) - \mu^2 = \frac{35}{12} \Rightarrow \sigma \approx 1.71$$

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Exercise

9.5.10 (supp) Two independent experiments are performed.

$P(\text{1st experiment succeeds}) = 0.7$

$P(\text{2nd experiment succeeds}) = 0.2$

Random variable X counts the number of successful experiments.

(a) Expected value of X ? $E(X) = 0.7 + 0.2 = 0.9$

(b) Probability of exactly one success? $0.7 \cdot 0.8 + 0.3 \cdot 0.2 = 0.62$

(c) Probability of at most one success? (b) + $0.3 \cdot 0.8 = 0.86$

(e) Variance of X ? $\sigma^2 = (0.62 \cdot 1 + 0.14 \cdot 4) - 0.9^2 = 0.37$

Exercise

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Cumulative Distribution Functions

Definition

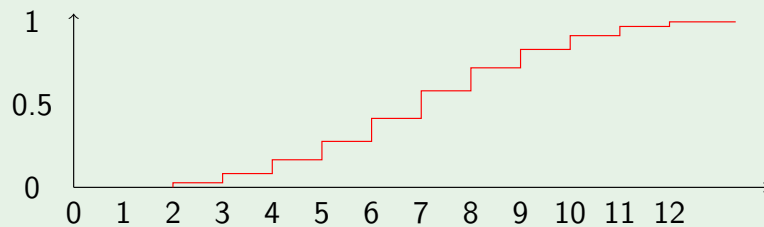
The **cumulative distribution function** $\text{CDF}_X : \mathbb{Z} \rightarrow \mathbb{R}$ of an integer random variable X is defined as

$$\text{CDF}_X(y) \mapsto \sum_{k \leq y} P(X = k)$$

$\text{CDF}_X(y)$ collects the probabilities $P(X)$ for all values up to y

Example

Cumulative distribution function for sum of 2 dice



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Example: Binomial Distributions

Definition

Binomial random variables count the number of 'successes' in n independent experiments with probability p for each experiment.

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$$

$$\text{CDF}_B(y) \mapsto \sum_{k \leq y} \binom{n}{k} p^k (1 - p)^{n-k}$$

Theorem

If X is a binomially distributed random variable based on n and p , then $E(X) = n \cdot p$ with variance $\sigma^2 = n \cdot p \cdot (1 - p)$

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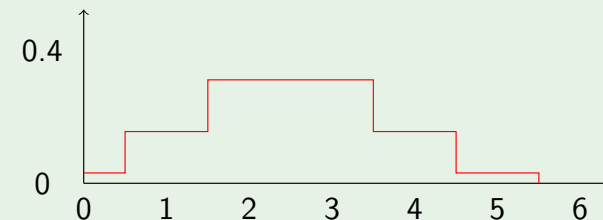
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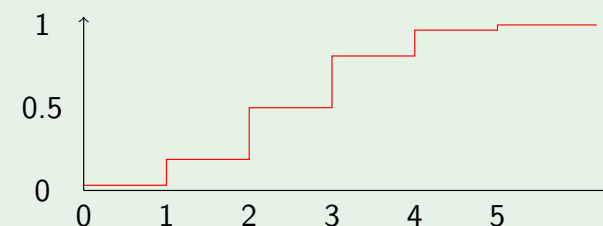
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Example (binomial distribution)

No. of HEADS in 5 coin tosses



CDF for no. of HEADS in 5 coin tosses



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Exercise

9.4.10 An experiment is repeated 30,000 times with probability of success $\frac{1}{4}$ each time.

(a) Expected number of successes? $E(X) = 30,000 \cdot \frac{1}{4} = 7500$

(b) Standard deviation? $\sigma = \sqrt{30,000 \cdot \frac{1}{4} \cdot \frac{3}{4}} = 75$

Exercise

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Normal Distribution

Fact

For large n , binomial distributions can be approximated by **normal distributions** (a.k.a. **Gaussian distributions**) with mean $\mu = n \cdot p$ and variance $\sigma^2 = n \cdot p \cdot (1 - p)$



$$\frac{1}{\sqrt{2\sigma^2\pi}} \cdot e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

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Summary

- Random variables X
- Expected value $E(X)$
- Mean μ , CDF, standard deviation σ , variance σ^2

Coming up ...

- Exam
- Holidays

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Course Review

Goal: for you to become a competent computer **scientist**.

Requires an understanding of fundamental concepts:

- number-, set-, relation- and graph theory
- logic and proofs, recursion and induction
- order of growth of functions
- combinatorics and probability

In CS/CE these are used to:

- formalise problem specifications and requirements
- develop abstract solutions (algorithms)
- analyse and prove properties of your programs

Examples:

- The University Course Timetabling Problem (→ [PDF](#))
- COMP9801 (Extended Design and Analysis of Algorithms)

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Course Review

- COMP9024 – Data Structures and Algorithms (23T3)

Concept	Used for
logic and proofs	correctness of algorithms
properties of relations	reachability in graphs
graphs	shortest path problems
trees	search trees
$\mathcal{O}$ (big-Oh)	efficiency of algorithms & data structures
alphabets and words	string algorithms
probability, expectation	randomised algorithms

NB

"universitas" (Lat.) = sum of all things, a whole

By acquiring knowledge and enhancing your problem-solving skills,
you're preparing yourself for the future

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Assessment Summary

- 1 quizzes — max. mark 14
- 2 mid-term assignment — max. mark 26
- 3 final exam — max. mark 60

NB

- $\text{QuizMark} = \max \left\{ \text{quizzes}, \text{ExamMark} * \frac{14}{60} \right\}$
- $\text{MidtermMark} = \max \left\{ \text{mid-term}, \text{ExamMark} * \frac{26}{60} \right\}$

NB

To pass the course, the sum of your marks

$$\text{Sum} = \text{QuizMark} + \text{MidtermMark} + \text{ExamMark}$$

must be 50 or higher **and** your ExamMark must be 25 or higher.

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Final Exam

Check your marks with classrun; Example:

FinalExam: 45/60
MidtermAsst: 19.5/26
Quiz: 11.3/14
Sum: 75.8

midterm: 17/26
quizzes: 11.3/14

Note: $\max\{17, 45 * \frac{26}{60}\} = 19.5$

Goal: to check whether you are a competent computer scientist.

Requires you to demonstrate:

- understanding of mathematical concepts
- ability to apply these concepts and explain how they work

Lectures, study of problem sets and quizzes have built you up to this point.

[Prac Exams](#) on course webpage (→ Practice exams)

Final Exam

2 hour (+10 mins reading time) online test
Wednesday, 3 May between 1:00pm and 3:15pm local time

NB

You must start the exam between 1:00pm and 1:05pm in order to get the full 130 minutes (= 2 hours 10 mins)

Format:

- Covers **all** of the contents of this course
- 8 numerical/short answer/multiple-choice (with ≥ 1 correct), each worth 4 marks
- 4 open questions, each worth 7 marks
- Maximum marks: $4*8 + 28 = 60$

If you

... are uncertain about how to interpret a question
... are unsure about how to answer a question
... find a question too difficult

⇒ do answer to the best of your understanding
⇒ do focus on the questions that you find easier
⇒ do not agonise about a question or your answer after you've submitted

Revision Strategy

- Re-read lecture slides
- Read the corresponding chapters in the book (R & W)
- **Review/solve problem sets**
- Solve more problems from the book
- Attempt prac exams (paper-based) on course webpage

NB

Applying mathematical concepts to solve problems is a skill that improves with practice

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Fit To Sit

If you attend an exam

- you declare that you are “fit to do so”
- it is your only chance to pass (i.e. no second chances)

If during an exam you are unwell and can't continue

- stop working, take note of the time
- immediately apply for Special Consideration

NB

If you experience a technical issue:

- Take screenshots of as many of the following as possible:
error messages, screen not loading, timestamped speed tests, power outage maps
- If issue was severe, apply for Special Consideration after conclusion of exam. Attach screenshots.



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Assessment

Assessment is about determining how well you understand the syllabus of this course.

If you can't demonstrate your understanding, you don't pass.

In particular, I can't pass people just because ...

- please, please, ... my family/friends will be ashamed of me
- please, please, ... I tried really hard in this course
- please, please, ... I'll be excluded if I fail COMP9020
- please, please, ... this is my final course to graduate
- etc. etc.

(Failure is a fact of life. For example, my scientific papers or project proposals get rejected sometimes too)

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Assessment (cont'd)

Of course, assessment isn't a “one-way street” ...

- I get to assess you in the final exam
- you get to assess me in UNSW's MyExperience Evaluation
 - go to <https://myexperience.unsw.edu.au/>
 - login using zID@ad.unsw.edu.au and your zPass

Response rate (as of Monday): < 50% 🥵

Please fill it out ...

- give me some feedback on how you might like the course to run in the future
- even if that is “Exactly the same. It was perfect this time.”

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The aim was for you to become a better computer scientist

- more confident in your own ability to use formal methods
- with a set of mathematical tools to draw on
- able to choose the right tool and analyse/justify your choices
- ultimately, enjoying solving problems in computer science

T h a t ' s A l l F o l k s

**Good Luck with the exam
and with your future computing studies**

